

A New Approach for Volumetric Knitting

François Guimbretière
Cornell University
Ithaca, New York, USA
francois@cs.cornell.edu

Victor Guimbretière
Ithaca High School
Ithaca, New York, USA
vg3@icsd.k12.ny.us

Amritansh Kwatra
Cornell Tech
New York, New York, USA
ak2244@cornell.edu

Scott E. Hudson
Carnegie Mellon University
Pittsburgh, Pennsylvania, USA
scott.hudson@cs.cmu.edu

ABSTRACT

Unlike 3D printers, which offer the ability to fabricate nearly arbitrary geometric forms, most textile fabrication processes are limited to the creation of sheets or hollow surface-based forms. This poster presents a new machine architecture to directly produce volumetric (solid 3D) knitted forms using a 2D bed of knitting needles, rather than the 1D line of needles used in conventional knitting. We describe a small prototype with 4x4 needles, and demonstrate that it can create fully volumetric knits, including overhangs.

CCS CONCEPTS

• Human-centered computing → Human computer interaction (HCI).

KEYWORDS

Knitting machines, additive manufacturing, 3D printing

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1 INTRODUCTION

While most knitting machines have been optimized to create surface-based forms [6, 8, 9, 11], in this poster, we consider a new approach to machine knitting which is volumetric. Knitting is performed at the top of the 2D array of needles and the form is pushed downward as the knitting progresses. This approach is related to a recent push to create volumetric textile forms [1, 5, 10, 12–14], and is most closely related to Hirose’s solid knitting machine [2–4, 7] which can create prisms. Yet, because we are using a bed of needles, our approach offers more freedom of connection within the volume and allows for a variety of new micro-structures arising from stitch-to-stitch connections to be fabricated. Figure 1 shows a 4x4 prototype used to validate our approach.

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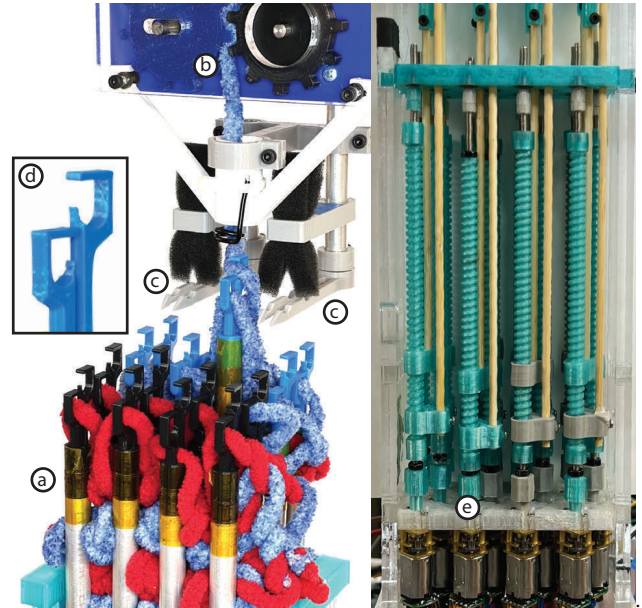


Figure 1: Our prototype in the process of assembling a 3D form. Our proof of concept implementation includes a 4x4 bed of needles (a), a yarn feeder (b) and two loop transfer tools (c). Each needle includes two hooks (d), that are independently actuated using 3D printed lead screws (e). As the fabrication proceeds, the form is pushed downward.

2 PRINCIPLE OF OPERATION

As illustrated in Figure 1d, our machine uses a unique new form of double hook *compound* (or *slide*) needle along with a new type of *loop transfer tool* (Figure 1c) and an active yarn feeder (Figure 1b). Performing a basic stitch proceeds as follows: We begin the stitch from a standard configuration with an existing loop held inside the closed front hook (Figure 2a). As a first step, the back hook catches a new segment of yarn from the yarn feeder mechanism moving past it, to form a new loop (Figure 2b), and the stem of the front hook is raised to close the hook. The loop transfer tool is then used to help move the existing loop over the new loop (Figure 2c,d).

The loop transfer tool is used again to tighten the loop by holding the previous loop down (Figure 2e). Finally, to restore the standard configuration, the transfer tool is used to transfer the loop from

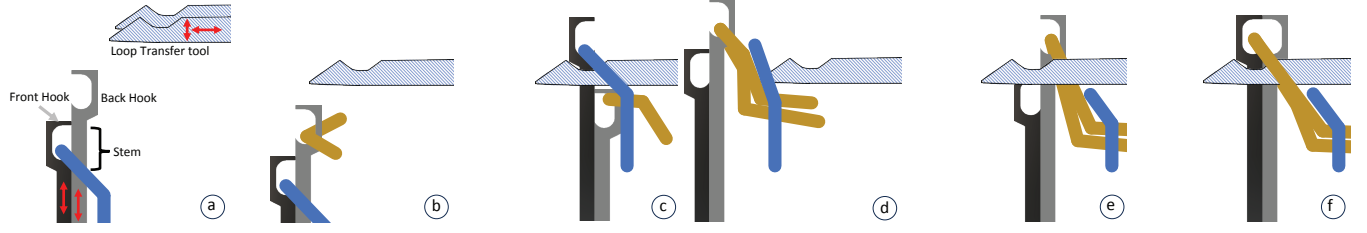


Figure 2: Steps for a simple purl stitch using our approach.

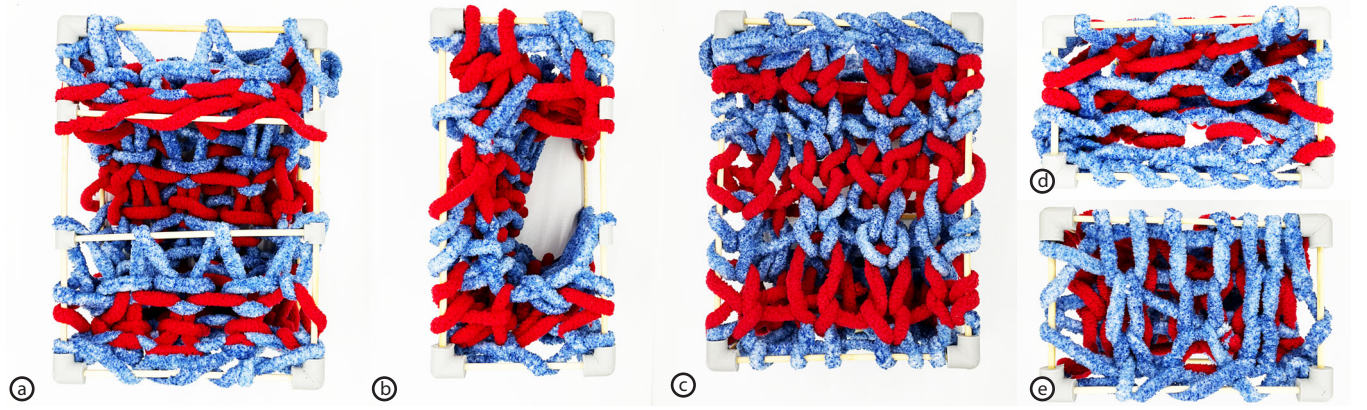


Figure 3: A $\square$ shape form (a = back view, b = side view, c = front view) was created on a 4x4 needle configuration, using alternating colors for each layer. It is displayed on a frame of 7cm(D) x 12cm(W) x 16.5cm(H). Each segment was created by connecting horizontally knitted layers and adding vertical stitches on the back and front. After creating the blue 4x4 base in horizontal knit (e) and continuing with two additional 4x4 layers, we added two 4x2 layers, and finished with two 4x4 layers, casting off at the top (d). Note the evidence of diagonal yarn transfers during horizontal knitting seen in (e).

the back hook to the front hook (Figure 2f). The back hook is then raised to return the needle to the standard configuration (Figure 2a) where the loop is held until the next operation. Our design can also perform cast-on and cast off.

Conventional knitting using a linear row of needles forms a nominally flat fabric below the bed of needles, and in our context can be considered *vertical* knitting. In addition to this type of row over row knitting, in our volumetric knitting approach, we also have the capability to knit *horizontally* across the 2D bed of needles. This horizontal knitting forms a 2D layer, rather than a 1D row. Creating a horizontal stitch is similar to a basic stitch, but the original loop is kept in the same needle and the newly created loop is transferred to a nearby destination needle. Note that for horizontal knitting, stitches can be connected to neighbors in all three directions, not just left or right along a row. This new ability opens up new possibilities for the creation of functional properties, such as anisotropic elasticity (e.g., by making connections front to back but not right to left, or vice versa), and engineered stiffness through selective overstretching and/or creation of patterned voids within a solid.

Figure 3 shows a fully volumetric knit with a $\square$ profile created on our 4x4 configuration. The base of the form was constructed using three horizontal knitted layers connected together, adding vertical knits on the back and front row to form a single volume.

We then dropped the loop of the two back rows (blue needles in Figure 1); then knitted the two 4x2 horizontal knitted layers connected together (and to the base), again adding vertical knits, on the back and front rows. Finally, we capped the volume with two more full 4x4 layers before dropping the loops of the three back rows and casting off the loops in the front row.

3 CONCLUSION AND ACKNOWLEDGEMENTS

The new knitting architecture presented in this paper creates volumes rather than only planar or surface-based objects. Our approach adds greater flexibility of possible connections between stitches. As a result, a substantial number of new micro-structures with different micro-level material properties can be created. These, in turn, should facilitate a range of new functional patterns that can be embedded in the resulting solid, but soft, objects.

Our proof-of-concept machine offers the opportunity to explore these interesting possibilities. However, it should be noted that it has many limitations and cannot run unassisted. The small 4x4 bed of our machine is clearly limiting, but it could easily be extended to larger settings. We believe that the current prototype provides initial evidence for the feasibility of this new approach to volumetric knitting and provides a solid basis to explore a new space of applications for knitted objects that has been opened up with our design. This work was supported in part by NSF grant IIS-1422106.

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